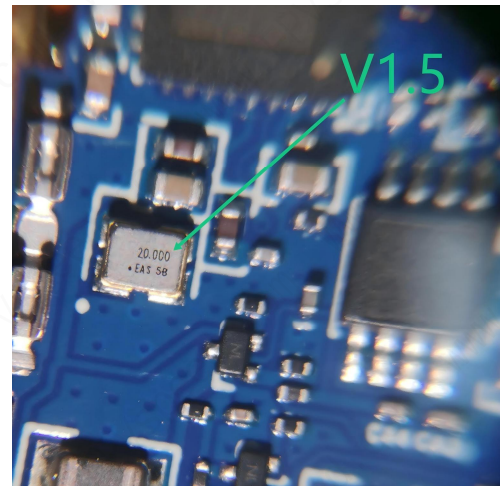
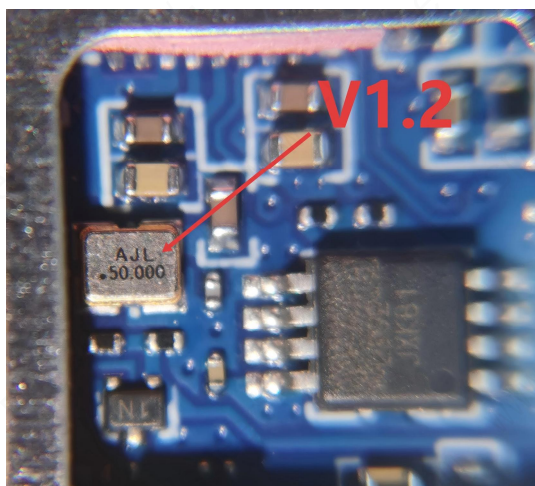
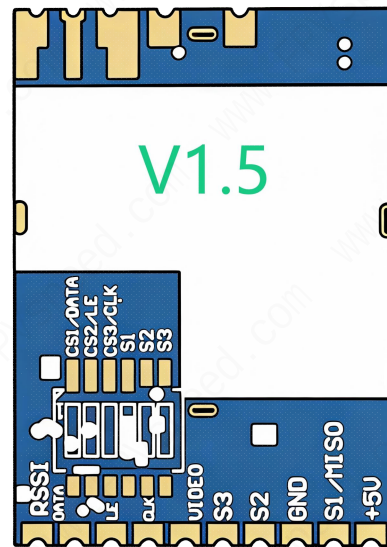
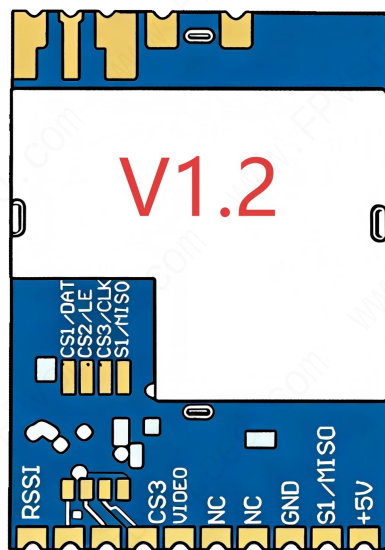


Q: How to distinguish FS3138RX between V1.2 and V1.5 versions?

A: As shown in the figure below



Q: How to run V1.2 50MHz code on your 20MHz crystal V1.5 hardware?

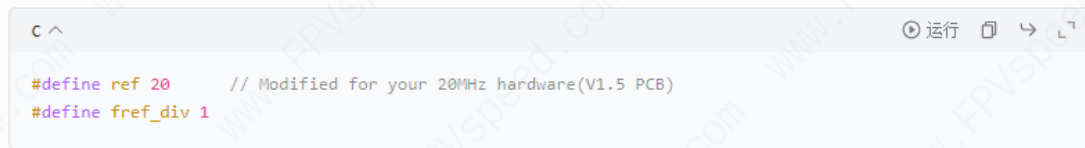
A: Please find below the simple code revision instruction when you use **V1.2 original source code on your V1.5 hardware which is equipped with 20MHz reference crystal oscillator** (V1.2 default setting is for 50MHz crystal).

Required Modification (Only 1 line change in PLL.h, NO changes needed for any .c source files)

1. Open header file: PLL.h
2. Locate the macro definition below (V1.2 default configuration):

```
c ^  
#define ref 50      // Default for V1.2(50MHz crystal)  
#define fref_div 1
```

3. Revise the ref value from 50 to 20 to match your onboard 20MHz crystal, keep fref_div = 1 unchanged:

A screenshot of a code editor window. The title bar shows a 'C' icon and a caret. The editor contains two lines of C preprocessor code:

```
#define ref 20 // Modified for your 20MHz hardware(V1.5 PCB)
#define fref_div 1
```

 The code is color-coded: `#define` is purple, `ref` is orange, `20` is pink, `//` is green, `Modified for your 20MHz hardware(V1.5 PCB)` is grey, `fref_div` is purple, and `1` is pink. The window has standard icons for running, saving, and navigating.

Important Notes

1. All other macros, SPI driver codes, frequency lookup table `Tab_PLL_Freq[]`, PLL register calculation logic in `PLL.c` remain untouched after modification.
2. After code revision, compile & download the project directly; the RF output frequency(2690~2990MHz defined in frequency table) will keep consistent with original design.
3. Your hardware circuit stays unchanged, no PCB/component modification required.
4. Loop filter components on PCB are optimized for 20MHz PFD frequency(V1.5 design), so no need to replace RC filter parts.

Please let us know if you encounter any compiling or PLL lock issues after modification.